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I'm a student of IIT Madras BS in Data Science. I'm interested in full-stack development.

Description

The Hospital Management System is a web-based application designed to simplify and automate the management of hospital operations. The system provides a platform where administrators, doctors, and patients can interact efficiently to manage appointments, medical records, and treatment history.

AI/LLM Used: Approximately **30% overall**

This includes:

Backend (Flask + SQLAlchemy+celery+redis):

Used for routing, models, database integration, and logic.

Frontend (vue + Bootstrap): Handled manually.

All the code was written and tested by me. LLM assistance was used only for debugging and clarifying errors , not for complete code generation.

Technologies Used

- Flask for AP(JWT Extender)
- VueJS for UI
- Jinja2 templates if using CDN **only for entry point**
- Bootstrap for HTML generation and styling
- SQLite for database Redis for caching
- Redis and Celery for batch jobs

DB Schema Design

User

id, username, email, password, role, is_active

Doctor

id, user_id (FK), specialization, department_id

Patient

id, user_id (FK)

Department

id, name

Appointment

id, doctor_id (FK), patient_id (FK), date, time, status, diagnosis, treatment, prescription, **doctor_notes**

Availability

id, doctor_id (FK), date, time, is_available

Relationships:

User ↔ Doctor: One-to-One

User ↔ Patient: One-to-One

Doctor ↔ Appointment: One-to-Many

Patient ↔ Appointment: One-to-Many

Doctor ↔ Availability: One-to-Many

Department ↔ Doctor: One-to-Many

Implemented Features:-

Role-based authentication and authorization (Admin / Doctor / Patient)

Admin dashboard to manage doctors, patients, departments and appointments

Doctor dashboard to manage availability, approve/reject appointments and add treatment details

Patient dashboard to view doctors, book appointments and manage bookings

Doctor availability management with date and time slots

Appointment booking system with status updates (Booked → Approved → Completed / Rejected)

Patient medical history tracking including diagnosis, treatment and prescriptions

Doctor access to full patient medical history for consultation

Search functionality for doctors by name or specialization

Search functionality for patients by name or ID for admin

CSV export of patient treatment history using background task

Daily appointment reminders using scheduled Celery jobs

Monthly doctor activity reports generated in HTML and sent via email

Caching implemented for frequently accessed APIs to improve performance

Video :

[video link](#)